TAITIEN

PXW-3

Crystal Oscillator Programmer

Take control of your oscillator supply — not your lead time.

Wrong frequency? Need one more? Don't wait 3–6 weeks. Program it yourself — today.

Purpose-built programming parameters co-developed with the oscillator's internal architecture — not reverse-engineered from a third-party tool.



The Programmer Built by the Oscillator Maker

PXW-3 is a USB-powered crystal oscillator programmer that configures CMOS oscillators to any frequency between 1 MHz and 200 MHz — on the bench, in under 2 seconds. When a frequency change or an added oscillator means waiting 3–6 weeks, design schedules stall. PXW-3 eliminates that dependency — program the exact frequency you need, when you need it, without placing a new order. Designed, validated, and supported by the same engineering team that manufactures the oscillators. Its built-in programming log retains a complete record of every session — frequency, timestamp, and batch — so every programmed device is fully traceable, from prototype to production.

Applications

- **Design Validation & Frequency Iteration —** instantly test and adjust frequencies without waiting for new samples
- **Last-Minute Design Changes & Risk Recovery —** fix frequency errors without delaying your project
- **NPI Builds & Low-Volume Production —** program exact quantities without MOQ or lead time constraints
- **On-Site Maintenance & Field Updates —** reconfigure oscillator frequency on-site, no factory return required

Key Features

› ≤ 2 seconds programming — no interruption to your workflow

› 1 MHz to 200 MHz — full frequency flexibility

› 5 SMD package sizes — one platform, multiple designs

› USB-powered — true bench-top operation

› Built-in traceability log — ready for production tracking

› Windows 7 and later

› On-demand programming — no factory lead time required

Why PXW-3?

Take control of your oscillator supply.

› No lead time dependency

› No re-order delays

› No frequency mismatch risk

Technical Specifications

Parameter	Specification
Frequency Range	1 MHz – 200 MHz
Max Frequency @ 1.8 V VCC	125 MHz
Oscillator Supply Voltage	1.8 V – 3.3 V (USB powered)
Programming Time	≤ 2 seconds per device (excl. post-program readback)
Communication & Power Interface	Mini USB
Operating System	Windows 7 or later
Supported Package Sizes	7.5×5.0 / 5.0×3.2 / 3.2×2.5 / 2.5×2.0 / 2.0×1.6 mm
Programmer Dimensions (L×W×H)	118 mm × 70 mm × 25 mm
Programmer Weight	121.8 g

Kit Contents

- PXW-3 CMOS Crystal Oscillator Programmer unit
- USB-A (Dual) to Mini-USB cable
- Programming socket 2.5 mm × 2.0 mm (PN: 1201-12040003) × 1
- Blank programmable crystal oscillator PY-002-B (2.5 mm × 2.0 mm) × 10 pcs

Additional sockets and blank oscillators for other package sizes are available separately — see Ordering Information.

One Source. Full Accountability.

- No interoperability gap — electrical validation is performed on the same production blanks, not on reference samples from a third-party tool vendor.
- Unified technical support — the same FAE team that handles oscillator selection supports the PXW-3. No boundary to navigate between programmer vendor and oscillator vendor.
- Roadmap alignment — new oscillator packages and socket compatibility are planned in parallel, not retrofitted after the fact.

Socket & Blank Oscillator Options

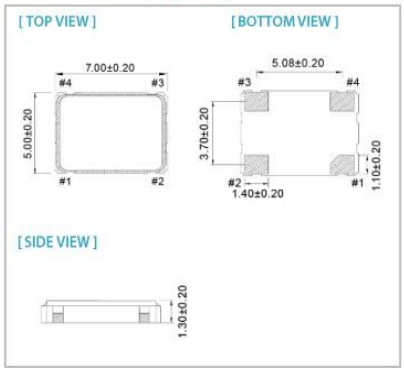
Sockets and blanks are ordered separately by package size. The kit ships with the 2.5×2.0 mm socket by default.

Package Size	Socket Orderable PN	Blank Oscillator PN
7.5 mm × 5.0 mm	1201-18010065	PC-002-B
5.0 mm × 3.2 mm	1201-18010066	PV-002-B
3.2 mm × 2.5 mm	1201-12040002	PX-002-B
2.5 mm × 2.0 mm	1201-12040003	PY-002-B
2.0 mm × 1.6 mm	1201-18010068	PZ-001-B

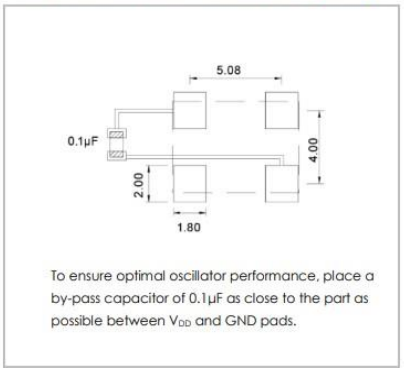
Blank Oscillator Mechanical Drawings

7.5 mm × 5.0 mm | PC-002-B

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)

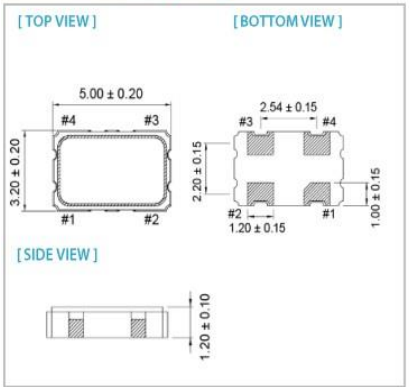


PIN FUNCTION (mm)

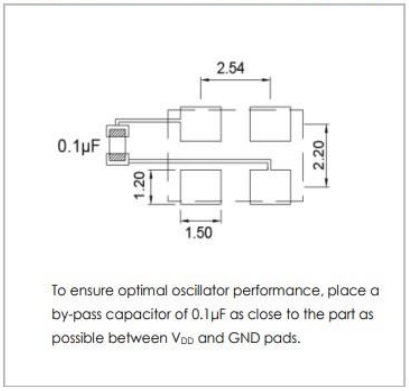
PIN#	FUNCTION
1	Tri-State
2	GND
3	Output
4	V_{DD}

5.0 mm × 3.2 mm | PV-002-B

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)

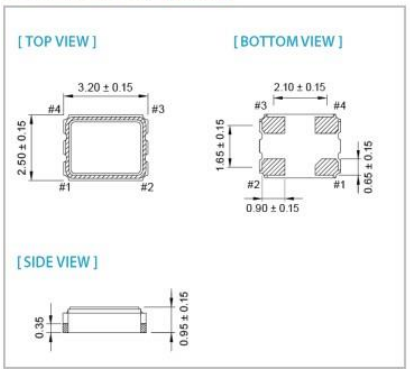


PIN FUNCTION (mm)

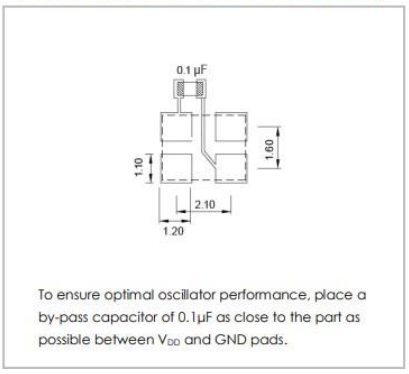
PIN#	FUNCTION
1	Tri-State
2	GND
3	Output
4	V_{DD}

3.2 mm × 2.5 mm | PX-002-B

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)



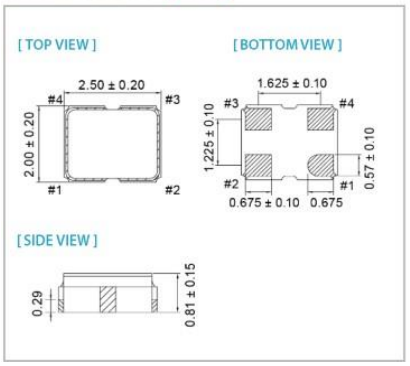
To ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between V_{DD} and GND pads.

PIN FUNCTION (mm)

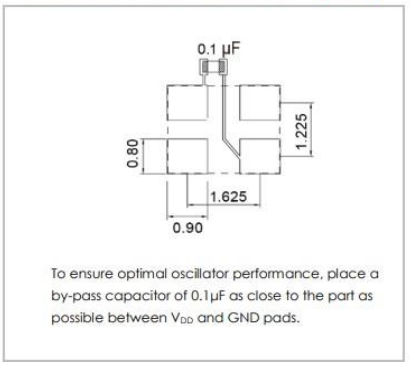
PIN#	FUNCTION
1	Tri-State
2	GND
3	Output
4	V_{DD}

2.5 mm × 2.0 mm | PY-002-B

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)



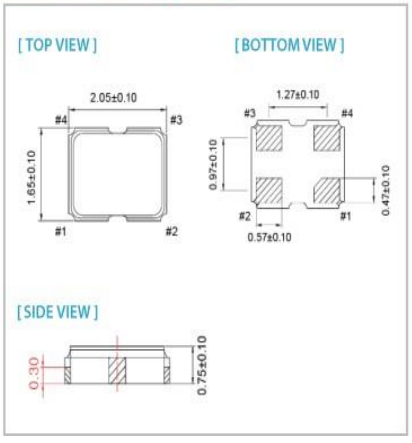
To ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between V_{DD} and GND pads.

PIN FUNCTION (mm)

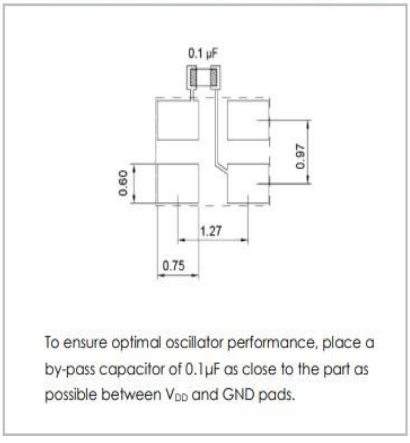
PIN#	FUNCTION
1	Tri-State
2	GND
3	Output
4	V_{DD}

2.0 mm × 1.6 mm | PZ-001-B

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)



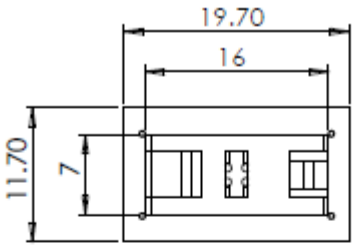
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PIN FUNCTION (mm)

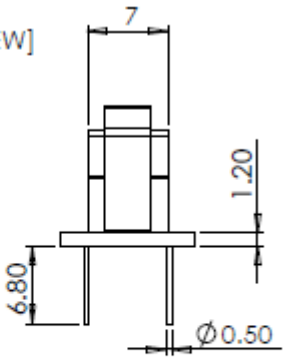
PIN#	FUNCTION
1	Tri-State
2	GND
3	Output
4	V_{DD}

Socket Mechanical Drawings

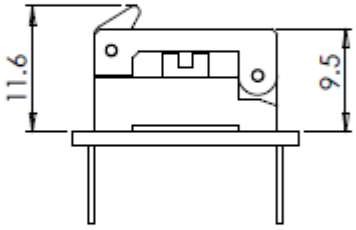
[TOP VIEW]

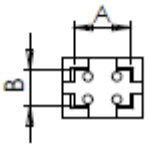


[SIDE VIEW]

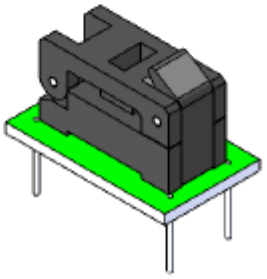


[FRONT VIEW]





[3D VIEW]



Part Number	Oscillator size	A	B
PC-002-B	7.5 mm × 5.0 mm	$7.0^{+0.3}_{+0.2}$	$5.0^{+0.3}_{+0.2}$
PV-002-B	5.0 mm × 3.2 mm	$5.0^{+0.3}_{+0.2}$	$3.2^{+0.3}_{+0.2}$
PX-002-B	3.2 mm × 2.5 mm	$3.2^{+0.15}_{+0.1}$	$2.5^{+0.15}_{+0.1}$
PY-002-B	2.5 mm × 2.0 mm	$2.5^{+0.15}_{+0.1}$	$2.0^{+0.15}_{+0.1}$
PZ-001-B	2.0 mm × 1.6 mm	$2.05^{+0.15}_{+0.1}$	$1.65^{+0.15}_{+0.1}$

Ordering Information

The PXW-3 Kit includes the programmer unit, USB cable, 2.5×2.0 mm socket, and 10× PY-002-B blank oscillators. To program other package sizes, order the corresponding socket and blanks separately.

Part Number	Description	Kit Contents
PXW-3 Kit	PXW-3 Programmer Kit (2.5×2.0 mm socket included)	Programmer + USB Cable + Socket + 10× PY-002-B
Additional Sockets		
1201-18010065	Socket, 7.5 mm × 5.0 mm	—
1201-18010066	Socket, 5.0 mm × 3.2 mm	—
1201-12040002	Socket, 3.2 mm × 2.5 mm	—
1201-12040003	Socket, 2.5 mm × 2.0 mm	—
1201-18010068	Socket, 2.0 mm × 1.6 mm	—
Blank Oscillators		
PC-002-B	Blank Oscillator, 7.5×5.0 mm	—
PV-002-B	Blank Oscillator, 5.0×3.2 mm	—
PX-002-B	Blank Oscillator, 3.2×2.5 mm	—
PY-002-B	Blank Oscillator, 2.5×2.0 mm	—
PZ-001-B	Blank Oscillator, 2.0×1.6 mm	—

Contact sales@taitien.com.tw or your regional distributor for volume pricing, blank minimum quantity, and extended socket options.

Ready to Get Started?

Stop waiting on oscillator lead times. Take control of your supply — today.

From frequency change to verified device — in seconds, not weeks.

To learn more about PXW-3, visit **www.taitien.com**
For sales enquiries, contact **sales@taitien.com.tw**

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